

Excel for Data Analysis — Corporate Interview Question Bank

Dataset Link: [Dataset Interview](#)

● QUESTION 1 — Understanding the Dataset & Data Quality

Difficulty: ★ Beginner → Intermediate

Type: Conceptual + Analytical Thinking

Scenario

You have joined an e-commerce company as a Junior Data Analyst.

Your manager gives you the Sales_Transactions sheet and asks:

"Before you build any dashboard, tell me whether this data is reliable enough for management reporting."

The dataset contains fields such as:

- Order_ID
- Order_Date
- Customer_ID
- Region_Raw
- Product_Raw
- Category_Raw
- Quantity
- Unit_Price
- Discount_Pct
- Sales_Amount
- Profit
- Order_Status
- Rating

Tech With Shivansh
Learn Skills • Build Projects • Get Hired

Interview Questions

1. What would you check before using this dataset for reporting?
2. Which columns would you consider critical for validating the dataset?
3. How would you identify duplicate transactions?
4. Why can duplicate Order_ID values be dangerous?
5. Would you immediately delete duplicate records? Why or why not?
6. What would you check in Sales_Amount and Profit?
7. How would you determine whether missing values are actually a problem?
8. What is the difference between a **data-quality issue** and a **business exception**?
9. If the dataset contains 1,000 rows, does that automatically mean there are 1,000 valid transactions?
10. What checks would you perform before presenting any KPI to management?

🎯 What the interviewer is testing

Whether you understand that **analysis starts with data validation, not charts**.

● QUESTION 2 — Raw Data vs Clean Data

Difficulty: ★ ★ Intermediate

Type: Conceptual + Power Query/Excel Thinking

Scenario

While reviewing the dataset, you notice that the Region_Raw column contains variations such as:

- North
- north
- NORTH
- North
- blank

Similarly, Payment_Method_Raw contains inconsistent capitalization and spacing.

Your manager asks:

"Can we simply use this column in a Pivot Table?"

Questions

1. What problem can inconsistent text create in analysis?
2. Would Excel treat North, north, and NORTH as the same category in every analytical operation?
3. Why is standardization important before aggregation?
4. How would inconsistent values affect a regional sales report?
5. Would you clean this manually or use Power Query? Explain your decision.
6. When should raw columns be preserved instead of overwritten?
7. What is the difference between **cleaning data** and **changing business data**?
8. If you find a blank region, would you automatically replace it with "Unknown"?
9. What would you investigate before making that replacement?
10. How would you validate that your cleaning process didn't change legitimate business values?

Corporate expectation

A good analyst doesn't just say "I'll use TRIM."

They explain **why the inconsistency exists, how it affects aggregation, how to standardize it, and how to validate the transformation.**

QUESTION 3 — Revenue Calculation Logic

Difficulty: ★★ Intermediate

Type: Formula Logic + Business Reasoning

Scenario

A sales manager asks:

"How is Sales Amount calculated in this dataset?"

You have:

- Quantity
- Unit_Price
- Discount_Pct
- Sales_Amount

For example:

Quantity	Unit Price	Discount
5	₹10,000	10%

Questions

1. How should the gross sales value be calculated?
2. How should the discount be incorporated?
3. What would happen if Discount_Pct is blank?



4. What if discount is stored as 10 instead of 10%?
5. How would you determine whether the existing Sales_Amount is correct?
6. Should you overwrite an existing sales amount if your calculated value differs?
7. What possible business reasons could explain a difference?
8. How would you identify unusually large or small sales transactions?
9. Would you use a formula, Pivot Table, Power Query or DAX for this validation?
10. How would you communicate a discovered calculation inconsistency to your manager?

Critical Thinking

The interviewer is looking for someone who understands:

Formula $\neq$ Truth.

A formula can be mathematically correct while the underlying business definition is wrong.

● QUESTION 4 — Profit vs Revenue

Difficulty: ★ ★ Intermediate

Type: Business Understanding + Logical Thinking

Scenario

Your manager says:

"The company generated ₹50 lakh in sales this month, so the month was successful."

You disagree because the dataset also contains:

- Cost_Per_Unit
- Quantity
- Sales_Amount
- Profit
- Discount_Pct

Questions

1. Why is Sales Amount alone insufficient to evaluate business performance?
2. How would you conceptually calculate profit?
3. Can a high-sales product have low profit?
4. Can a low-sales product generate a high profit margin?
5. How can excessive discounting affect profit?
6. If Region A has higher sales than Region B but lower profit, what would you investigate?
7. Which KPI would you prioritize for a profitability discussion?
8. Why should an analyst distinguish between **profit amount** and **profit margin**?
9. What business decisions could be made incorrectly by looking only at revenue?
10. If management asks you to identify the "best region," what additional information would you request before answering?

Interview Trap

There is **no universally correct "best region" based only on sales.**

The definition of "best" depends on the business objective.

● QUESTION 5 — Order Status and KPI Definition

Difficulty: ★★ Intermediate

Type: Business Logic + Critical Thinking

Scenario

The Order_Status column contains:

- Completed
- Cancelled
- Returned
- Pending

The management dashboard currently shows:

Total Sales = SUM(Sales_Amount)

The Finance Manager challenges the number:

"Why are cancelled and returned orders included in our sales KPI?"

Questions

1. Should every row necessarily contribute to Total Sales?
2. What is the difference between **gross sales**, **net sales**, and **realized sales**?
3. Should cancelled orders contribute to revenue?
4. Should returned orders be treated differently from cancelled orders?
5. Should pending orders be included?
6. How would you define a management-ready Sales KPI?
7. Would your answer depend on the company's accounting policy?
8. What questions would you ask the business before building the KPI?
9. How could the wrong definition create misleading management decisions?
10. How would you document the definition of the KPI so another analyst gets the same result?

🎯 **What makes this a corporate question**

The interviewer isn't asking:

"Can you SUM a column?"

They're asking:

"Can you define what should actually be summed?"

That's a much more important analyst skill.

● QUESTION 6 — Customer Analysis

Difficulty: ★★ Intermediate

Type: Analytical Thinking + Excel Logic

Scenario

The company wants to understand customer performance.

The dataset contains:

- Customer_ID
- Customer_Name
- Customer_Segment
- Region
- Sales_Amount
- Quantity
- Order_Status

The Head of Sales asks:

"Who are our most valuable customers?"

Questions

1. What does "most valuable customer" actually mean?
2. Would you rank customers by Sales Amount alone?
3. Would number of orders matter?
4. Would profit matter?
5. Would customer segment matter?
6. How would you distinguish between:
 - High-value customer
 - High-frequency customer
 - High-volume customer
 - High-profit customer
7. How could one unusually large transaction distort your ranking?
8. What additional KPI would you calculate alongside customer sales?
9. If a customer generates ₹10 lakh revenue but only ₹20,000 profit, would you call them valuable?
10. How would you present your conclusion to the Sales Manager?

Strong Analyst Thinking

"Top customer" is not a metric.

You need to define the metric first.

● QUESTION 7 — Regional Performance Investigation

Difficulty: ★★☆☆ Intermediate → Advanced

Type: Critical Thinking

Scenario

Management gives you this requirement:

"Find our worst-performing region and tell us why."

You have:

- Region
- State
- City
- Sales
- Profit
- Quantity
- Orders
- Customer Segment
- Sales Channel

Questions

1. What metric would you use to identify the worst-performing region?
2. Why might Sales alone produce the wrong conclusion?
3. What if one region has low sales but extremely high profit margin?
4. What if one region has high sales but high returns?
5. How would you drill from:

Region → State → City → Customer/Product



6. What would you investigate after identifying an underperforming region?
7. Would you compare absolute values or percentages?
8. Why can percentage metrics sometimes be misleading?
9. How would you handle a region with very few transactions?
10. What evidence would you require before telling management **why** the region is underperforming?

🔥 Interviewer's Hidden Test

The correct answer is **not simply "create a Pivot Table by Region."**

The real test is whether you can move from:

What happened? → Where? → Why? → What should the business do?

🟠 QUESTION 8 — Pivot Table Decision-Making

Difficulty: ★★☆☆ Intermediate

Type: Excel + Analytical Design

Scenario

Your manager wants a quick management report showing:

- Total Sales
- Total Profit
- Quantity Sold
- Number of Orders
- Sales by Region
- Sales by Category
- Monthly Sales Trend

You are given the raw transaction table.

Questions

1. Would you build separate Pivot Tables or one large Pivot Table?
2. Which fields belong in **Rows**?
3. Which fields belong in **Values**?
4. Which fields would you potentially use as **Filters/Slicers**?
5. Would you use COUNT, COUNTA, or DISTINCT COUNT for orders?
6. Why could counting rows give an incorrect number of orders in some datasets?
7. What chart would you choose for monthly sales?
8. What chart would you choose for regional comparison?
9. Would a pie chart be appropriate for every categorical analysis?
10. How would you prevent a Pivot Table from becoming a collection of numbers with no business meaning?

🎯 Interview Expectation

You should be able to explain **why you chose the Pivot structure**, not just how to create one.

🔴 QUESTION 9 — Dashboard KPI Design

Difficulty: ★★☆☆ Advanced

Type: Critical Thinking + Business Analytics

Scenario

You are asked to build an executive sales dashboard.

The CEO has **10 seconds** to look at the dashboard before entering a meeting.

You can display only **5 primary KPIs**.

Available metrics include:

- Total Sales
- Total Profit
- Profit Margin
- Total Quantity
- Number of Orders
- Number of Customers
- Average Order Value
- Return Rate
- Cancellation Rate
- Customer Rating

Questions

1. Which 5 KPIs would you select?
2. Why?
3. Which KPI would you place first?
4. Would you show Sales and Profit together?
5. Would you include Quantity?
6. Why might Profit Margin be more useful than Profit alone?
7. Would Average Order Value be useful?
8. Would Return Rate belong on an executive dashboard?
9. Should every KPI have a target or benchmark?
10. What makes a KPI **actionable** rather than merely interesting?

🔥 Advanced Follow-up

The CEO says:

"Sales increased by 15%, but profit decreased by 8%. Explain."

What dimensions would you investigate first?

Possible directions:

Region → Product → Discount → Channel → Customer Segment → Order Status

Explain **why you would investigate them in that order**.

🔴 QUESTION 10 — The Executive Challenge

Difficulty: ★★☆☆ Advanced

Type: Corporate Case Interview + Critical Thinking

Scenario

You are presenting the monthly sales report to the Sales Director.

Your report shows:

Sales ↑ 18%

Quantity ↑ 11%

Profit ↓ 4%

The Sales Director says:

"This doesn't make sense. If sales increased by 18%, why did profit fall?"

You have access to the entire dataset.



Your Task

Do **not** immediately calculate random metrics.

Explain your investigation strategy.

Questions

1. What would you check first?
2. Could discounting explain the difference?
3. Could product mix explain it?
4. Could cost differences explain it?
5. Could the increase in sales come from low-margin products?
6. Could region mix be responsible?
7. Could channel mix be responsible?
8. Could returned/cancelled transactions distort the result?
9. Could data-quality issues cause the apparent decline?
10. What analysis would you perform to identify the root cause?
11. What Excel features would you use?
12. Would you use formulas, Pivot Tables, Power Query, Power Pivot or some combination?
13. How would you validate your conclusion?
14. What would you show the Sales Director?
15. Most importantly:

What business recommendation would you make after finding the root cause?

Expected Analyst Thinking

Your investigation should eventually move toward something like:

Sales Increase

↓

What drove the increase?

↓

Product / Region / Channel / Customer

↓

Was the growth profitable?

↓

Discount + Cost + Product Mix

↓

Identify Root Cause

↓

Recommend Action

Difficulty Progression — First 10

Q	Area	Difficulty	Primary Skill
1	Data Quality	★	Data validation
2	Data Cleaning	★★	Standardization
3	Revenue Logic	★★★	Formula reasoning
4	Profitability	★★★	Business understanding
5	KPI Definition	★★★	Business logic
6	Customer Analysis	★★★	Metric design



7	Regional Analysis	★ ★ ★	Root-cause analysis
8	Pivot Tables	★ ★ ★	Excel analytical design
9	Dashboard KPIs	★ ★ ★	Executive reporting
10	Executive Case	★ ★ ★ ★	End-to-end analytical thinking



Tech With Shivansh
 Learn Skills • Build Projects • Get Hired

● QUESTION 11 — Lookup Strategy: Which Function Would You Choose?

Difficulty: ★ ★

Type: Formula + Conceptual + Corporate Scenario

Scenario

You are given three tables:

Sales_Transactions

- Order_ID
- Customer_ID



<https://www.linkedin.com/in/the-venom/>

Shivansh

- Product_ID
- Quantity
- Sales_Amount
- Profit

Customer_Dim

- Customer_ID
- Customer_Name
- Region
- State
- City
- Customer_Segment

Product_Dim

- Product_ID
- Product
- Category
- Standard_Price
- Cost_Per_Unit

Your manager asks:

"Add Customer Region and Product Category to the transaction report."

Interview Questions

1. Which Excel function would you choose for this requirement?
2. Would you use VLOOKUP, XLOOKUP, INDEX + MATCH, or Power Query?
3. Why?
4. What happens if the Customer_ID doesn't exist in the Customer table?
5. How would you avoid showing #N/A?
6. Would IFERROR() be appropriate here?
7. What is the difference between hiding an error and fixing the underlying data problem?
8. If the lookup table contains duplicate Customer_IDs, what problem does that create?
9. How would you validate that the lookup returned the correct customer?
10. If this operation must be repeated every month, would you continue using formulas?

🔥 Follow-up

The interviewer says:

"The dataset has 1 million rows. Would your answer change?"

Explain your choice based on **performance, maintainability and refreshability**, not personal preference.

🟡 QUESTION 12 — XLOOKUP vs VLOOKUP

Difficulty: ★★

Type: Technical Interview

Scenario

Your company currently uses:

`=VLOOKUP(C2, Customer_Dim!A:F, 3, FALSE)`

to retrieve Region.

A new analyst suggests replacing it with:

=XLOOKUP(C2, Customer_Dim!A:A, Customer_Dim!C:C, "Not Found")

Questions

1. Explain what both formulas are doing.
2. What limitations does traditional VLOOKUP have?
3. Why is XLOOKUP generally more flexible?
4. Can VLOOKUP look to the left?
5. Can XLOOKUP return a value from a column positioned before the lookup column?
6. Why is an exact-match lookup important for IDs?
7. What happens if multiple matching IDs exist?
8. Why should "Not Found" sometimes be preferred over IFERROR()?
9. Would you replace every existing VLOOKUP with XLOOKUP automatically?
10. What compatibility consideration would you check before using XLOOKUP in a corporate environment?

Interview Test

The interviewer wants to know whether you understand **function capability + business environment**, not whether you can recite that XLOOKUP is "better."

● QUESTION 13 — SUMIFS for Business Analysis

Difficulty: ★★☆☆

Type: Formula + Business Scenario

Scenario

The Sales Director asks:

"How much completed sales did we generate from the North region for Electronics?"

You need to consider:

- Region_Raw
- Category_Raw
- Order_Status
- Sales_Amount

Task

Design an Excel formula that calculates:

Completed + North + Electronics Sales

Questions

1. Which function would you use?
2. Why is SUMIFS() appropriate?
3. What would be your criteria ranges?
4. How would you structure the formula?
5. What happens if the category contains "electronics" instead of "Electronics"?
6. What if the region contains leading/trailing spaces?
7. Should you clean the data first or compensate for the inconsistency inside the formula?
8. What if the manager wants the calculation to change dynamically based on cells containing Region and Category?
9. How would you convert the formula into a reusable reporting component?
10. How would you validate the result against a Pivot Table?

Follow-up



<https://www.linkedin.com/in/the-venom/>

Shivansh

Now the manager asks:

"Give me completed sales for North **OR** South."

Explain why this is different from a normal multi-criteria SUMIFS() problem.

● QUESTION 14 — COUNTIFS and Business Conditions

Difficulty: ★★☆☆

Type: Formula + Logical Thinking

Scenario

The Customer Success team asks:

"How many completed orders from Enterprise customers were generated through the Online channel?"

Relevant fields:

- Customer_Segment
- Order_Status
- Sales_Channel
- Order_ID

Questions

1. Which Excel function would you use?
2. How would you construct the logic?
3. What is the difference between COUNT, COUNTA, COUNTIF, and COUNTIFS?
4. Why might counting rows not always equal counting orders?
5. If Order_ID contains duplicates, what happens?
6. How would you identify whether duplicate IDs exist?
7. If management wants **unique orders**, would COUNTIFS() alone be sufficient?
8. What alternative Excel approaches could solve the unique-count problem?
9. How would you explain the difference between **transaction count** and **unique order count**?
10. Which definition should appear on the dashboard?

🔗 Corporate Thinking

A weak answer:

"Use COUNTIFS."

A strong answer:

"First I need to establish whether the business wants transaction rows or distinct orders, because duplicate Order_IDs can make a row count misleading."

● QUESTION 15 — AVERAGEIFS and Misleading Averages

Difficulty: ★★☆☆

Type: Statistical Thinking + Excel

Scenario

The management team wants:

"Average order value for Enterprise customers in the West region."

You have:

- Customer_Segment
- Region
- Sales_Amount



- Order_Status

Questions

1. Which Excel function would you use?
2. Should cancelled orders be included?
3. Should returned orders be included?
4. What exactly is the denominator in Average Order Value?
5. Is:

Total Sales ÷ Number of Orders

always equivalent to the average of Sales_Amount?

6. What happens if duplicate transactions exist?
7. How could extreme transactions affect the average?
8. When might median be more useful than average?
9. How would you investigate whether the average is representative?
10. If Enterprise AOV is ₹80,000 and Consumer AOV is ₹7,000, what business questions would you ask next?

🔥 Critical Thinking

The interviewer is testing whether you understand that **an average is a business metric with a definition**, not merely an AVERAGE() formula.

● QUESTION 16 — Nested IF / IFS: Customer Classification

Difficulty: ★★☆☆

Type: Logical Formula + Business Rules

Scenario

The Sales Manager wants customers classified based on their total sales:

Total Sales	Customer Tier
≥ ₹5,00,000	Platinum
≥ ₹2,00,000	Gold
≥ ₹50,000	Silver
< ₹50,000	Bronze

Questions

1. How would you implement this in Excel?
2. Would you use nested IF() or IFS()?
3. In what order should the conditions be evaluated?
4. What happens if you check < ₹50,000 first?
5. How would you handle blank sales?
6. How would you handle negative sales?
7. Should customer tier be calculated at transaction level or customer level?
8. Why is that distinction important?
9. How would you make the thresholds configurable rather than hard-coded?
10. Would you consider this a good candidate for Power Query or DAX instead of an Excel formula?

🔥 Follow-up

The Sales Head changes the thresholds every quarter.

Would you:



- A. Edit the formula manually?
- B. Store thresholds in a separate table?
- C. Use Power Query?
- D. Use Power Pivot/DAX?

Explain your choice.

● QUESTION 17 — Date Analysis: Monthly Sales

Difficulty: ★★ ★

Type: Date Functions + Business Analysis

Scenario

The Finance Manager asks:

"Show monthly sales from January 2024 to August 2026."

The dataset contains Order_Date.

Some dates may require cleaning/type correction before analysis.

Questions

1. How would you convert the transaction date into Month and Year?
2. Would you use MONTH() and YEAR() separately?
3. Would you create a Year-Month column?
4. Why is "Jan" alone insufficient when the dataset covers multiple years?
5. Why is January 2024 different from January 2025?
6. How would you sort months chronologically rather than alphabetically?
7. Would you use TEXT(Order_Date, "MMM-YYYY")?
8. What problem can arise when using text-based month labels?
9. Would a Pivot Table or formula-based report be better for this requirement?
10. Why is a proper Date Dimension useful when working with Power Pivot/DAX?

🎯 Expected Thinking

The candidate should recognize:

Date → Year → Month → Time Period → Aggregation

rather than treating dates as ordinary text.

● QUESTION 18 — NETWORKDAYS / Business-Day Analysis

Difficulty: ★★ ★

Type: Date Functions + Real Business Scenario

Scenario

The Operations team wants to measure how quickly customer orders are processed.

Suppose you later receive:

- Order_Date
- Dispatch_Date
- Delivery_Date

The manager asks:

"Calculate the number of working days between Order Date and Delivery Date."

Questions

1. Which Excel function would you consider?
2. Why would DATEDIF() or simple date subtraction potentially be inappropriate?



3. What is the difference between calendar days and business days?
4. How does NETWORKDAYS() treat weekends?
5. When would you use NETWORKDAYS.INTL()?
6. How would public holidays affect the calculation?
7. Where would you store the holiday list?
8. If Friday-to-Monday is considered, how many calendar days and working days are involved?
9. How would you identify orders exceeding the company's SLA?
10. How could you convert the result into an operational KPI?

🔥 Business Extension

Create the conceptual metric:

SLA Breach = Delivery Working Days > Target Working Days

Then explain how management could use it.

● QUESTION 19 — Data Cleaning Without Destroying Information

Difficulty: ★★☆☆

Type: Critical Thinking + Data Quality

Scenario

You discover these values in Region_Raw:

North
north
NORTH
North
North
North
(blank)

The manager says:

"Just replace everything with North and move on."

Questions

1. What would you do first?
2. How would you determine whether these are genuinely the same business category?
3. Which Excel functions could help identify the problem?
4. Where could TRIM() help?
5. Where could CLEAN() help?
6. Would UPPER() solve the entire problem?
7. How would you detect blank values?
8. Should blanks automatically become "Unknown"?
9. What if the blank region actually means the transaction was never assigned to a region?
10. How would you preserve the original data while creating a cleaned field?
11. Would you perform this cleaning in Excel formulas or Power Query?
12. How would you make the cleaning process repeatable when next month's data arrives?

🔥 Interview Challenge

The interviewer asks:

"Why shouldn't we simply clean the original column?"

Your answer should discuss:

Tech With Shivansh
Learn Skills • Build Projects • Get Hired



● QUESTION 20 — Build a Management Report from Raw Data

Difficulty: ★★☆☆

Type: End-to-End Corporate Case

Scenario

You have just joined the company as a Data Analyst.

Your manager gives you the Sales_Transactions workbook and says:

"I need a management-ready sales report by tomorrow morning."

The report must contain:

Executive KPIs

- Total Sales
- Total Profit
- Total Orders
- Total Customers
- Average Order Value
- Profit Margin

Analysis

- Sales by Region
- Profit by Region
- Sales by Category
- Sales by Channel
- Monthly Sales Trend
- Top 10 Products
- Top 10 Customers

Management Requirement

The report must be:

- Accurate
- Refreshable
- Easy to understand
- Scalable
- Reusable next month

Your Interview Task

Explain your complete approach **before touching Excel**.

Questions

1. What would you inspect first?
2. How would you validate the raw data?
3. How would you handle duplicate Order IDs?
4. How would you handle missing values?
5. How would you standardize inconsistent text?
6. Would you use formulas or Power Query for cleaning?
7. Would you create Pivot Tables?
8. Would you create a Data Model?
9. When would Power Pivot become useful?



10. Which KPIs would you calculate using formulas initially?
11. How would you validate your KPI calculations?
12. Which charts would you select for management?
13. What would you avoid putting on the dashboard?
14. How would you make the report refreshable?
15. How would you design the workbook so another analyst can maintain it?
16. What documentation would you provide?
17. How would you test the final report?
18. What would you do if your Total Sales doesn't match Finance's number?
19. How would you investigate the discrepancy?
20. How would you explain the final insights to a non-technical manager?

The Corporate-Level Answer Structure

For Question 20, a strong candidate should naturally think in this sequence:

RAW DATA

↓

DATA VALIDATION

↓

DATA CLEANING

↓

DATA TRANSFORMATION

↓

DATA MODEL

↓

CALCULATIONS

↓

PIVOT / ANALYSIS

↓

DASHBOARD

↓

VALIDATION

↓

BUSINESS INSIGHTS

↓

RECOMMENDATION

That is the level we're aiming for in this 50-question set.

Questions 11–20 Skill Coverage

Q	Main Area	Difficulty	Interview Skill
11	Lookup Strategy	★ ★	Tool selection
12	XLOOKUP vs VLOOKUP	★ ★	Technical concepts
13	SUMIFS	★ ★ ★	Conditional aggregation
14	COUNTIFS	★ ★ ★	Counting logic
15	AVERAGEIFS	★ ★ ★	Metric interpretation

Tech With Shivansh
Learn Skills • Build Projects • Get Hired

16	IF / IFS	☆☆☆	Business rules
17	Date Analysis	☆☆☆	Time-based analysis
18	NETWORKDAYS	☆☆☆	Operational analytics
19	Data Cleaning	☆☆☆☆	Data-quality reasoning
20	End-to-End Report	☆☆☆☆	Corporate case study

● QUESTION 21 — Pivot Table: Regional Sales Investigation

Difficulty: ☆☆☆

Type: Pivot Table + Business Analysis

Scenario

The Sales Head asks:

"Give me the performance of every region and identify where management attention is required."

You have:

- Region_Raw
- State
- City
- Sales_Amount
- Profit
- Quantity
- Order_Status
- Sales_Channel

Task

Build a Pivot-based analysis containing:

- Total Sales by Region
- Total Profit by Region
- Quantity by Region
- Order Count by Region
- Average Sales per Order

Interview Questions

1. Which field would you place in the Rows area?



2. Which fields belong in Values?
3. How would you calculate Average Sales per Order?
4. Would you use Sum of Sales_Amount or Average of Sales_Amount?
5. How would you exclude cancelled transactions?
6. Would you use a filter or clean the data first?
7. How would you compare regions fairly if transaction volumes differ significantly?
8. Which metric would you use to identify the weakest region?
9. What if one region has the highest sales but lowest profit margin?
10. What additional drill-down would you perform before making a recommendation?

🔥 Follow-up

The manager says:

"Don't just tell me which region is worst. Tell me **why**."

Explain how you would drill:

Region → State → City → Product → Channel → Customer Segment

🟠 QUESTION 22 — Pivot Table: Top Products

Difficulty: ★★ ★

Type: Pivot Table + Ranking

Scenario

The Product Manager asks:

"Which products are driving our business?"

You need to identify the **Top 5 products by Sales**.

Questions

1. Which fields would you use in your Pivot Table?
2. How would you sort products from highest to lowest sales?
3. How would you filter to the Top 5?
4. Would you rank by Sales, Profit, Quantity or Margin?
5. Why can the Top 5 by Sales differ from the Top 5 by Profit?
6. What if a product has extremely high sales but negative profitability?
7. Would you recommend discontinuing it immediately?
8. What additional dimensions would you investigate?
9. How would you compare product performance across regions?
10. How could a Pivot Chart communicate the result effectively?

🎯 Critical Thinking

If the CEO asks:

"What's our best product?"

Don't answer immediately.

First clarify:

Best by what?

- Revenue?
- Profit?
- Margin?
- Quantity?
- Customer adoption?



- Growth?

● QUESTION 23 — Pivot Chart Selection

Difficulty: ★★ ★

Type: Visualization + Business Communication

Scenario

You need to present the following:

Requirement A

Sales by Region

Requirement B

Monthly Sales Trend

Requirement C

Sales by Category

Requirement D

Profit by Product

Requirement E

Sales Channel Contribution

Questions

Choose an appropriate chart for each requirement and justify your choice.

1. Which chart would you use for regional comparison?
2. Which chart would you use for monthly trends?
3. Which chart would you use for category comparison?
4. Which chart would you use for Top 10 products?
5. Would you use a pie chart for Sales Channel?
6. When does a pie/doughnut chart become difficult to interpret?
7. Why is a line chart appropriate for time-series analysis?
8. Why can a 3D chart be problematic?
9. Should every Pivot Table have a corresponding chart?
10. What makes a chart useful for decision-making rather than decorative?

🔥 Interview Test

The interviewer gives you **10 different charts** and asks:

"Which one would you remove from the executive dashboard?"

Your answer must be based on **information density, clarity and decision usefulness**, not personal preference.

● QUESTION 24 — Conditional Formatting: Find the Problem

Difficulty: ★★ ★

Type: Excel + Analytical Thinking

Scenario

Your manager asks you to highlight:

- Negative Profit
- High Discount
- Low Rating
- High Sales



- Duplicate Order IDs

Questions

1. Which conditional formatting rules would you use?
2. Would you use built-in rules or formula-based rules?
3. How would you highlight negative Profit?
4. How would you identify duplicate Order IDs?
5. How would you highlight transactions where Discount > 15%?
6. How would you identify high-value orders?
7. How could conditional formatting help with data-quality checking?
8. Why should excessive conditional formatting be avoided?
9. How would you make the thresholds dynamic?
10. Can conditional formatting replace proper data analysis?

🔥 Scenario Extension

Management says:

"Highlight transactions that are both high discount AND low profit."

Explain how you would build the logical condition.

Conceptually:

Discount > Threshold

AND

Profit < Threshold

🔴 QUESTION 25 — Data Validation: Preventing Bad Data

Difficulty: ★★☆☆

Type: Data Quality + Excel Controls

Scenario

You are responsible for an Excel template used by 20 sales executives.

They manually enter:

- Region
- Product
- Sales Channel
- Payment Method
- Quantity
- Discount
- Rating

Management complains that users enter:

North

north

NORTH

Nort

North Region

Questions

1. How would you prevent this problem?
2. How can Data Validation help?
3. Would you allow free-text entry for Region?



4. Where would you store the approved list?
5. How would you restrict Discount to valid values?
6. How would you restrict Rating to 1-5?
7. How would you prevent negative Quantity?
8. What is the difference between **preventing bad data** and **cleaning bad data later**?
9. Which approach is better for recurring operational files?
10. What happens if the validation list itself changes?

Corporate Thinking

A mature analyst doesn't only ask:

"How do I clean bad data?"

They ask:

"How do I stop bad data from entering the system in the first place?"

QUESTION 26 — Dynamic Arrays: FILTER()

Difficulty: ★★★★★

Type: Advanced Formula + Dynamic Analysis

Scenario

The Sales Director wants a live report showing only:

Completed orders from the North region with Sales above ₹50,000.

The report should automatically update when the source data changes.

Questions

1. Would FILTER() be suitable?
2. What conditions would you define?
3. How would you combine multiple conditions?
4. What happens when no records match?
5. How would you prevent an ugly error when there are no results?
6. What is a spilled array?
7. Why is FILTER() different from manually applying an Excel filter?
8. What happens when something blocks the spill range?
9. How could you make the Region and Sales threshold user-controlled?
10. Would you use this approach for a recurring corporate report containing hundreds of thousands of rows?

Advanced Follow-up

The user selects:

Region = West

Minimum Sales = ₹75,000

Your output should automatically display matching transactions.

Explain how you would design this without manually changing the formula every time.

QUESTION 27 — UNIQUE() + SORT()

Difficulty: ★★★★★

Type: Dynamic Arrays + Logical Thinking

Scenario

The HR/Operations team wants a dynamically generated list of:

All unique employees involved in sales transactions.

They also want the list sorted alphabetically.

Questions

1. Which function would return unique values?
2. Which function would sort the result?
3. How would you combine them?
4. What happens when a new employee appears in the dataset?
5. Why is a dynamic array better than manually maintaining the list?
6. How could UNIQUE() help with data-quality investigation?
7. How could you use UNIQUE() to find unique regions?
8. How could you use it to investigate inconsistent categories?
9. What happens if "North" and "north" are treated differently in your source data?
10. Would you clean the data before generating the unique list?

🔥 Analyst Scenario

You generate a unique Region list and unexpectedly get:

North

north

NORTH

North

What does this tell you about your data?

And what should you do **before** using the list for reporting?

🔴 QUESTION 28 — Two-Way Lookup

Difficulty: ★★★★★

Type: INDEX + MATCH + Analytical Logic

Scenario

A management pricing matrix contains:

Product	Online	Retail	Corporate
Laptop	...	...	...
Monitor	...	...	...
Smartphone	...	...	...

The user selects:

- Product
- Sales Channel

Excel must return the corresponding value.

Questions

1. What is a two-way lookup?
2. Why might INDEX + MATCH be useful?
3. How would you dynamically identify the row?
4. How would you dynamically identify the column?
5. Why is hard-coding a column number dangerous?
6. Could XLOOKUP() solve this problem?
7. How would you handle an invalid Product/Channel combination?
8. What if the matrix structure changes?



9. When would you prefer a normalized data structure instead of a lookup matrix?
10. What does this question reveal about the difference between **Excel skills and data-modeling skills**?

Interview Insight

Knowing how to perform a lookup is useful.

Knowing **when the data structure itself should be redesigned** is more valuable.

QUESTION 29 — Dashboard: KPI vs Insight

Difficulty: ★★★★★

Type: Dashboard Design + Critical Thinking

Scenario

You build an executive dashboard containing:

- Total Sales
- Total Profit
- Total Orders
- Customers
- Quantity
- Sales by Region
- Sales by Category
- Monthly Trend
- Top Products

The CEO looks at it and says:

"I can see the numbers. But what am I supposed to do?"

Questions

1. What is missing from the dashboard?
2. What's the difference between a KPI and an insight?
3. Should a dashboard only report historical numbers?
4. How would you identify an underperforming region?
5. How would you highlight a significant change?
6. Should KPIs have targets?
7. What is the purpose of variance?
8. How would you show actual vs target?
9. How could conditional formatting improve executive interpretation?
10. How would you turn:

"West Region sales are ₹12 lakh"
into a meaningful business insight?

Strong Answer

A number describes **what happened**.

An insight explains:

What happened → Why it happened → Why it matters → What should happen next

QUESTION 30 — The 10-Second Executive Dashboard Test

Difficulty: ★★★★★

Type: Corporate Case Study + Critical Thinking

Scenario

You are given exactly **10 seconds** to communicate the company's current performance to the CEO.

Your available metrics are:

- Total Sales
- Total Profit
- Profit Margin
- Orders
- Customers
- Quantity
- Average Order Value
- Return Rate
- Cancellation Rate
- Customer Rating
- Sales by Region
- Sales by Category
- Monthly Sales Trend
- Top Products

The CEO will not read a table containing 50 numbers.

Task

Design the conceptual structure of your dashboard.

Interview Questions

1. Which 5 KPIs would you place at the top?

Explain each selection.

2. What would you put directly below the KPI cards?

Why?

3. Which chart would communicate monthly performance?

Why?

4. How would you identify the region requiring attention?

5. How would you distinguish a sales problem from a profitability problem?

6. What if Sales increased 20% but Profit decreased 5%?

What would you investigate?

7. What if Profit increased but Customers decreased?

What could explain it?

8. What if AOV increased while Order Count decreased?

What hypotheses would you test?

9. How would you make the dashboard interactive?

Potential tools:

- Slicers
- Timeline
- Pivot Tables
- Pivot Charts
- Dynamic formulas

10. How would you make sure the dashboard remains useful when the dataset grows from:

1,000 → 100,000 → 1,000,000 rows?



🧠 The Real Interview Challenge

The interviewer then gives you this statement:

"Sales are up 18%. Good job."

Your job is to **challenge the statement**.

Ask:

Revenue

Why did Sales increase?

↓

Volume

Did Quantity increase?

↓

Orders

Did order volume increase?

↓

Pricing

Did average selling price increase?

↓

Discount

Did discounts change?

↓

Product Mix

Did customers buy higher-value products?

↓

Geography

Which regions drove the growth?

↓

Channel

Which sales channels drove it?

↓

Profitability

Did Profit and Margin also increase?

↓

Quality

Are the numbers reliable?

↓

Decision

What should management do next?

That is the difference between **Excel reporting** and **Excel-based data analysis**.

🇮🇳 Questions 21–30 Skill Coverage

Q	Area	Difficulty	Primary Skill
21	Pivot Analysis	★ ★ ★	Business segmentation
22	Top Products	★ ★ ★	Ranking & interpretation
23	Pivot Charts	★ ★ ★	Visualization
24	Conditional Formatting	★ ★ ★	Exception detection

Tech With Shivansh
Learn Skills • Build Projects • Get Hired



25	Data Validation	★ ★ ★	Data governance
26	FILTER	★ ★ ★ ★	Dynamic analysis
27	UNIQUE + SORT	★ ★ ★ ★	Dynamic arrays
28	Two-Way Lookup	★ ★ ★ ★	Advanced formulas
29	Dashboard Insights	★ ★ ★ ★	Executive communication
30	Executive Dashboard	★ ★ ★ ★ ★	End-to-end reasoning

● QUESTION 31 — Power Query: Why Not Clean Data Manually?

Difficulty: ★ ★ ★

Type: Power Query + Corporate Scenario

Scenario

Every month, Finance sends you a new Excel file containing sales transactions.

The structure is almost identical every month, but you must:

- Remove duplicates
- Remove blank rows
- Fix data types
- Standardize Region
- Clean Product names
- Remove unnecessary columns
- Filter invalid transactions
- Load the cleaned data into Excel

Your manager says:

"You've already cleaned January's file. Just repeat the same steps for February."

Questions

1. Would you repeat the cleaning manually?
2. Why is Power Query more appropriate here?
3. What is the difference between **one-time cleaning** and **repeatable transformation**?
4. What happens to the Applied Steps when the next month's data arrives?
5. Why is refreshability important?
6. What happens if the source file structure changes?
7. Which transformations should ideally happen before reporting?
8. Why should raw data generally be kept separate from transformed output?
9. How does Power Query reduce human error?
10. At what point would you consider this workflow an automation problem rather than an Excel-formula problem?

🎯 Corporate Expectation

The correct mindset:

Don't automate because automation is fashionable. Automate because the same transformation is repeatedly required.

● QUESTION 32 — Power Query: Null vs Zero

Difficulty: ★ ★ ★

Type: Data Quality + Critical Thinking

Scenario

While cleaning the dataset, you find:

- Blank Rating
- Blank Region
- Quantity = 0
- Profit = 0

A junior analyst says:

"They're all basically zero. Replace all blanks with 0."

Questions

1. Is that logically correct?
2. What does null represent?
3. What does zero represent?
4. Why can replacing a missing value with zero distort analysis?
5. What could a missing Rating mean?
6. What could Quantity = 0 mean?
7. Would you remove all rows containing nulls?
8. When is replacing a null appropriate?
9. How would you decide whether a missing value is:
 - Unknown
 - Not applicable
 - Missing due to data-entry failure
 - Legitimately unavailable
10. How could incorrect null handling affect KPIs?

🔥 Interview Follow-Up

The interviewer asks:

"A customer's rating is blank. Can you calculate their average rating as zero?"

Explain why **missing** ≠ **zero**.

● QUESTION 33 — Append vs Merge

Difficulty: ★★☆☆

Type: Power Query + Conceptual

Scenario

You receive:

January Sales

Order ID	Product	Sales
----------	---------	-------

February Sales

Order ID	Product	Sales
----------	---------	-------

You also have:

Product Master

Product ID	Product	Category
------------	---------	----------

Questions

1. Would you **Append** January and February?
2. Would you **Merge** Sales with Product Master?
3. Explain the difference between Append and Merge.
4. What does Append conceptually do?



5. What does Merge conceptually do?
6. What must be true about the columns when appending?
7. What key would you use to merge Sales with Product Master?
8. What happens if the Product ID doesn't exist in Product Master?
9. Which operation resembles combining rows?
10. Which operation resembles a lookup/join?

Memory Rule

Append = Stack similar tables vertically

Merge = Bring related information horizontally

QUESTION 34 — Power Query Join Selection

Difficulty: ★★★★★

Type: Power Query + Data Modeling Logic

Scenario

You have:

Sales

100 products sold.

Product Master

120 products defined.

You want to identify:

"Which products have never been sold?"

Questions

1. Which Join Type would you use?
2. Why?
3. Would Inner Join solve the problem?
4. Would Left Outer Join solve it?
5. From which table should you start?
6. How would you identify unmatched records after the merge?
7. What would a Full Outer Join tell you?
8. When would an Inner Join be appropriate?
9. What is the difference between Left Outer and Right Outer Join?
10. Why is choosing the correct join direction important?

Corporate Follow-Up

The interviewer changes the requirement:

"Show me every product, including products with no sales."

What changes in your approach?

QUESTION 35 — Group By: From Transactions to Business Summary

Difficulty: ★★★★★

Type: Power Query + Analytical Thinking

Scenario

The raw dataset contains thousands of transactions.

Management wants:



Region	Total Sales	Total Profit	Order Count
--------	-------------	--------------	-------------

Questions

1. Would you use Group By?
2. What field would you group by?
3. Which aggregations would you calculate?
4. Why is Group By different from simply sorting the data?
5. What happens to transaction-level detail after aggregation?
6. When would you prefer a Pivot Table over Group By?
7. When would you prefer Power Query Group By?
8. Could Group By be part of an ETL process?
9. How would you create a summary by Region + Category?
10. How would you validate the grouped result against the original transaction data?

🎯 Advanced Thinking

If the business wants both:

Transaction-level data

and

Management summary

should you replace the raw table with the summary?

Why or why not?

🔴 QUESTION 36 — Power Pivot: Fact vs Dimension

Difficulty: ★★★★★

Type: Data Modeling + Corporate Interview

Scenario

Your workbook contains:

Sales_Transactions

- Order ID
- Date
- Customer ID
- Product ID
- Quantity
- Sales
- Profit

Customer_Dim

- Customer ID
- Customer Name
- Region
- State
- Segment

Product_Dim

- Product ID
- Product
- Category

Date_Dim

- Date
- Year
- Month
- Quarter

Questions

1. Which table is the Fact table?
2. Which tables are Dimensions?
3. Why shouldn't customer attributes be repeatedly stored in every transaction row?
4. What is a primary key?
5. What is a foreign key?
6. Which field connects Sales to Customer?
7. Which field connects Sales to Product?
8. Which field connects Sales to Date?
9. What problems can occur if the dimension key contains duplicates?
10. Why is this structure more scalable than a giant flat Excel table?

🎯 Interview Test

Explain this in one sentence:

"A fact table records business events; dimension tables provide the context used to analyze those events."

● QUESTION 37 — Relationship & Cardinality Problem

Difficulty: ★★★★★

Type: Power Pivot + Critical Thinking

Scenario

You create this relationship:

Customer_Dim[Customer_ID]

↓

Sales_Transactions[Customer_ID]

The relationship is expected to be:

One-to-Many

But Excel reports a relationship problem.

Questions

1. What should exist on the "one" side?
2. Why must Customer_ID be unique in Customer_Dim?
3. What happens if Customer_ID appears multiple times in Customer_Dim?
4. What happens if a Sales transaction references a customer that doesn't exist?
5. What is referential integrity?
6. Explain One-to-Many.
7. Explain Many-to-One.
8. When could Many-to-Many become problematic?
9. Why shouldn't you create relationships simply because two columns have similar names?
10. How would you troubleshoot an incorrect relationship?

🔥 Corporate Scenario

A manager says:

"Just remove duplicate customers until the relationship works."

Would you do that?

What would you investigate first?

● QUESTION 38 — DAX Measure vs Calculated Column

Difficulty: ★★★★★

Type: DAX + Analytical Design

Scenario

You need:

Total Sales

Total Profit

Profit Margin

Commission Amount

The Sales table contains 100,000 rows.

Questions

1. Which calculations would you consider Measures?
2. Which calculation might naturally be a Calculated Column?
3. What is a Measure?
4. What is a Calculated Column?
5. What is the difference between row-level and filter-context calculations?
6. Why can unnecessary calculated columns increase model size?
7. Why are Measures useful for dynamic reporting?
8. If a manager filters Region from North to West, should Total Sales recalculate?
9. Why would a Measure be appropriate?
10. Would you create a separate calculated column for every possible dashboard KPI?

🔥 Interview Challenge

The interviewer asks:

"If I can calculate Sales using an Excel formula, why do I need a DAX Measure?"

Give a business-focused answer rather than simply saying:

"Because DAX is more powerful."

● QUESTION 39 — VBA: Why Should an Analyst Learn It?

Difficulty: ★★★

Type: VBA Basics + Corporate Thinking

Scenario

Every morning, an analyst receives a report and performs the same sequence:

1. Open workbook
2. Remove duplicates
3. Delete blank rows
4. Format headers
5. Apply number formats
6. Resize columns
7. Create a summary
8. Save the final report



9. Send it to management

The process takes **30 minutes every day**.

Questions

1. Is this a good candidate for automation?
2. Why?
3. What is a Macro?
4. What is VBA?
5. What is the relationship between Macros and VBA?
6. How can Macro Recorder help a beginner?
7. Why might recording the process be easier than writing VBA from scratch?
8. What are the limitations of Macro Recorder?
9. When would VBA provide more flexibility than a recorded Macro?
10. How would you calculate the business value of automating this process?

🔥 Business Calculation

If automation saves:

30 minutes/day × 22 working days/month

how much analyst time is saved monthly?

Then ask:

Would that justify building and maintaining the automation?

🔴 QUESTION 40 — The Final VBA Automation Decision

Difficulty: ★★★★★

Type: Corporate Case + Decision Making

Scenario

Your manager says:

"We have a monthly Excel report. Every month someone spends 3 hours cleaning and formatting it. Can you automate it?"

The workflow involves:

- Removing duplicate rows
- Cleaning text
- Formatting columns
- Applying formulas
- Refreshing analysis
- Preparing a management-ready report

You know:

- Excel formulas
- Pivot Tables
- Power Query
- Power Pivot
- Basic DAX
- Macro Recorder
- Basic VBA

Interview Questions

1. Would you immediately build a VBA Macro?



<https://www.linkedin.com/in/the-venom/>

Shivansh

2. Which parts of the workflow would you consider for **Power Query**?
3. Which parts could be handled by **Pivot Tables / Power Pivot**?
4. Which calculations could be **DAX Measures**?
5. Which repetitive presentation tasks could potentially be handled by **VBA**?
6. What would you avoid automating?
7. How would you decide between Power Query and VBA?
8. How would you make the process refreshable?
9. How would you validate the automated output against the manual report?
10. What happens if the source file structure changes?
11. How would you document the automation?
12. How would you ensure another analyst can operate it?
13. What would you do if the automation takes longer to maintain than the manual process?
14. How would you measure the success of the automation?
15. What risks would you communicate to management?

🔥 THE REAL CORPORATE ANSWER

The best solution may **not** be:

"Build a VBA Macro."

A mature analyst might design:

Raw Files

↓

Power Query

Clean

Transform

Combine

↓

Power Pivot

Data Model

Relationships

↓

DAX

KPI Measures

↓

Pivot Tables / Charts

Analysis

↓

VBA

Optional report preparation / repetitive presentation automation

↓

Management Report

This is the important lesson:

Use the right Excel technology for the right problem. Don't force VBA into a problem Power Query already solves better.

Tech With Shivansh
Learn Skills • Build Projects • Get Hired

🚩 Questions 31-40 Skill Coverage



<https://www.linkedin.com/in/the-venom/>

Shivansh

Q	Area	Difficulty	Primary Skill
31	Power Query	★ ★ ★	ETL thinking
32	Null vs Zero	★ ★ ★	Data quality
33	Append vs Merge	★ ★ ★ ★	Transformation logic
34	Join Types	★ ★ ★ ★	Relational thinking
35	Group By	★ ★ ★ ★	Aggregation
36	Fact & Dimension	★ ★ ★ ★	Data modeling
37	Relationships	★ ★ ★ ★ ★	Model troubleshooting
38	DAX	★ ★ ★ ★ ★	Calculation design
39	VBA Basics	★ ★ ★	Automation thinking
40	VBA Case	★ ★ ★ ★ ★	Technology selection



Tech With Shivansh
 Learn Skills • Build Projects • Get Hired

● QUESTION 41 — Sales Increased, But Should Management Celebrate?

Difficulty: ★ ★ ★ ★ ★

Type: Critical Thinking + Business Analysis



<https://www.linkedin.com/in/the-venom/>

Shivansh

Scenario

You present the following monthly results:

KPI	Previous Month	Current Month
Sales	₹40 Lakh	₹48 Lakh
Orders	1,000	1,150
Quantity	2,400	2,650
Profit	₹8 Lakh	₹7.5 Lakh
Profit Margin	20%	15.6%

The Sales Head says:

"Excellent. Sales increased by 20%!"

The Finance Manager disagrees:

"I'm more concerned about profitability."

Questions

1. Who is correct?
2. Is the business necessarily performing better?
3. What happened to Profit?
4. What happened to Profit Margin?
5. What could cause Sales to increase while Profit decreases?
6. Could discounting explain this?
7. Could product mix explain this?
8. Could cost increases explain this?
9. Could channel mix explain this?
10. What analysis would you perform before recommending any action?

🔥 Interview Challenge

You have **2 minutes** to explain the situation to the CEO.

Give a concise answer covering:

What happened → Why → Business impact → Recommended action

🔴 QUESTION 42 — The KPI Doesn't Match Finance

Difficulty: ★★★★★

Type: Data Validation + Business Logic

Scenario

You calculate:

Total Sales = ₹52.4 Lakh

Finance reports:

Total Sales = ₹49.8 Lakh

Your manager asks:

"Why don't your numbers match?"

Questions

1. What would you investigate first?
2. Could cancelled orders be included in your calculation?
3. Could returned orders be included?
4. Could duplicate Order IDs exist?
5. Could date ranges differ?
6. Could Finance be using Net Sales rather than Gross Sales?
7. Could discounts be treated differently?
8. Could some transactions be excluded by business rules?
9. Could your Power Query transformation have changed the data?
10. How would you reconcile the two numbers?

🔥 Critical Rule

Never change your number simply because Finance has a different number.

First identify the **definition and calculation logic** behind both numbers.

Final Interview Question

"How would you prove which number is correct?"

🔴 QUESTION 43 — Data Quality Trap

Difficulty: ★★★★★

Type: Data Quality + Investigation

Scenario

Your dashboard shows:

Total Orders = 1,000

But after investigating Order_ID, you discover duplicate IDs.

Some orders appear more than once.

Your manager says:

"Remove duplicates and refresh the dashboard."

Questions

1. Would you immediately remove duplicate rows?



<https://www.linkedin.com/in/the-venom/>

Shivansh

2. What exactly constitutes a duplicate transaction?
3. Is duplicate Order_ID automatically evidence of duplicate data?
4. Could one Order ID legitimately have multiple transaction rows?
5. What other fields would you compare?
6. How would you investigate the duplicates?
7. What would happen if you removed legitimate line items?
8. How would you determine the correct grain of the dataset?
9. Why is **data grain** important?
10. How would you document your decision?

🔥 Advanced Question

Explain the difference between:

Duplicate Record

and

Multiple Rows Belonging to the Same Order

This distinction is critical in real-world analytics.

🔴 QUESTION 44 — Power Query vs Excel Formulas vs VBA

Difficulty: ★★★★★

Type: Tool Selection + Architecture

Scenario

Your monthly reporting workflow has these tasks:

1. Import 10 monthly files
2. Combine them
3. Remove duplicates
4. Clean text
5. Standardize categories
6. Calculate business metrics
7. Create Pivot Tables
8. Refresh dashboard
9. Format the final report
10. Save the report with a standard filename

Your manager asks:

"Which Excel technology should we use for each step?"

Your Task

Choose between:

- Excel Formulas



- Power Query
- Power Pivot
- DAX
- Pivot Tables
- VBA

Questions

Assign the most appropriate tool to each task and explain **why**.

🔥 Interview Trap

Don't answer:

"VBA can do everything."

Technically, VBA can automate many things.

But that doesn't mean it is the **best engineering choice**.

Strong reasoning should resemble:

Import + Combine → Power Query

Cleaning → Power Query

Data Model → Power Pivot

Business Measures → DAX

Interactive Analysis → Pivot Tables

Repetitive Presentation Automation → VBA

● QUESTION 45 — Dashboard Says "Sales Are Down"

Difficulty: ★★★★★

Type: Root-Cause Analysis

Scenario

Your dashboard shows:

▼ **Sales ↓ 12%**

The CEO asks:

"Why?"

You cannot simply answer:

"Because sales decreased."

That's describing the problem, not explaining it.

Questions

Build your investigation framework.

1. Compare current period vs previous period.
2. Check regional performance.
3. Check product/category performance.
4. Check sales channel.
5. Check customer segment.
6. Check order volume.
7. Check average order value.
8. Check discount percentage.
9. Check cancellations/returns.
10. Check whether the decline is caused by one major segment or widespread across the business.

🔥 Final Challenge

Suppose you discover:

- Orders ↓ 5%
- AOV ↓ 7%
- Electronics Sales ↓ 18%
- North Sales ↓ 22%
- Online Sales ↓ 15%

What would you investigate next?

Don't randomly explore everything.

Explain your **priority order and reasoning**.

🔴 QUESTION 46 — Build a KPI That Doesn't Lie

Difficulty: ★★★★★

Type: KPI Design + Business Logic

Scenario

Management wants:

"Customer Performance KPI"

Available fields:

- Customer ID
- Sales
- Profit



- Orders
- Quantity
- Rating
- Region
- Segment

One analyst proposes:

Customer KPI = Sales

Another proposes:

Customer KPI = Profit

Another proposes:

Customer KPI = Number of Orders

Questions

1. Who is correct?
2. Can all three metrics be useful?
3. What exactly should a "Customer Performance KPI" measure?
4. How would you define customer value?
5. Would you use Sales?
6. Would you use Profit?
7. Would you use Profit Margin?
8. Would Order Frequency matter?
9. Would customer retention matter?
10. What additional data would you request if the business wanted a serious Customer Lifetime Value analysis?

🔥 Interview Lesson

A KPI isn't automatically meaningful just because you can calculate it.

The business objective determines the KPI.

🔴 QUESTION 47 — Scalability Challenge

Difficulty: ★★★★★

Type: Architecture + Performance + Analyst Thinking

Scenario

Your current workbook contains:

1,000 transactions

It works perfectly.



<https://www.linkedin.com/in/the-venom/>

Shivansh

Your manager tells you:

"Next quarter, we'll have 1 million transactions."

The current workbook uses:

- Thousands of formulas
- Multiple VLOOKUPs
- Several Pivot Tables
- Manual cleaning
- Manual report formatting

Questions

1. Will your current approach scale?
2. What could become slow?
3. Which transformations should move to Power Query?
4. Why might a Data Model be better than a giant worksheet?
5. When would Power Pivot become useful?
6. Why are Measures preferable for many dynamic KPIs?
7. Would you keep every intermediate calculation as a worksheet column?
8. How would you reduce unnecessary calculations?
9. How would you make the process refreshable?
10. What architecture would you propose for the 1-million-row scenario?

🔥 Final Design

Design:

Source Files

→ Power Query

→ Data Model

→ DAX

→ Pivot Tables / Dashboard

→ Optional VBA Automation

Explain every arrow.

🔴 QUESTION 48 — The Business Insight Test

Difficulty: ★★★★★

Type: Business Storytelling + Critical Thinking

Scenario

Your analysis reveals:

West Region

Sales: ₹18 Lakh
Profit: ₹1.2 Lakh
Profit Margin: 6.7%

North Region

Sales: ₹14 Lakh
Profit: ₹3.5 Lakh
Profit Margin: 25%

The Sales Manager says:

"West is clearly our best region because it generates more revenue."

Questions

1. Do you agree?
2. Which region generates more Sales?
3. Which generates more Profit?
4. Which has better Profit Margin?
5. Why could West have higher sales but lower profitability?
6. What would you investigate?
7. Could discounting be responsible?
8. Could product mix be responsible?
9. Could customer segment be responsible?
10. What recommendation would you give management?

🔥 Final Challenge

Turn these numbers into a **2-3 sentence executive insight**.

Do not simply repeat the numbers.

Your answer should contain:

Observation + Explanation/Hypothesis + Action

🔴 QUESTION 49 — Full Excel Analyst Case Study

Difficulty: ★★★★★

Type: Complete Interview Simulation

Scenario

You join an organization as a Data Analyst.

Your manager gives you the provided **1,000-row Sales dataset** and says:

"Build a professional management reporting solution from this raw data."



You have one working day.

Requirements

Step 1 — Data Quality

Identify:

- Duplicate records
- Missing values
- Invalid values
- Inconsistent text
- Incorrect data types

Step 2 — Cleaning

Use appropriate Excel/Power Query techniques.

Step 3 — Data Model

Create:

Fact Sales

•

Customer Dimension

•

Product Dimension

•

Date Dimension

Step 4 — Analysis

Calculate:

- Total Sales
- Total Profit
- Orders
- Customers
- Quantity
- Average Order Value
- Profit Margin
- Regional Performance
- Category Performance
- Product Performance

Step 5 — Dashboard

Tech With Shivansh
Learn Skills • Build Projects • Get Hired



Create an executive dashboard.

Step 6 — Insights

Answer:

1. Which region performs best?
2. Which region requires attention?
3. Which products drive revenue?
4. Which products drive profit?
5. Which channel performs best?
6. Which customer segment contributes most?
7. What is the monthly trend?
8. Are discounts affecting profitability?
9. What anomalies exist?
10. What should management do next?

🔥 Interview Questions

The interviewer asks:

"Why did you choose each Excel feature?"

You must justify:

- Power Query
- Pivot Tables
- Power Pivot
- DAX
- Charts
- Conditional Formatting
- Data Validation
- Formulas

This tests whether you understand **solution design**, not just Excel syntax.

🔴 QUESTION 50 — FINAL MASTER INTERVIEW CASE

Difficulty: ★★★★★ +

Type: End-to-End Corporate Interview

Scenario

You are sitting in the final round for a **Data Analyst position**.

The interviewer gives you the Sales dataset.

They say:

"Don't show me a dashboard yet. First tell me how you would approach this problem."



You have **15 minutes** to explain your solution.

PART A — Understand the Business

1. What questions would you ask the stakeholder before touching the data?
 2. What is the business objective?
 3. What KPIs matter?
 4. Who will consume the report?
 5. How frequently will it be refreshed?
-

PART B — Understand the Data

6. What is the grain of the Sales table?
 7. What is a transaction?
 8. What is an Order?
 9. What are the Fact and Dimension tables?
 10. What keys connect them?
-

PART C — Validate

11. How would you check duplicates?
 12. How would you identify missing values?
 13. How would you identify inconsistent categories?
 14. How would you validate dates?
 15. How would you validate Sales and Profit calculations?
-

PART D — Transform

16. Which tasks belong in Power Query?
 17. Which transformations should be reusable?
 18. How would you handle new monthly files?
 19. How would you preserve the raw data?
 20. How would you make the workflow refreshable?
-

PART E — Model

21. Would you create a Data Model?
 22. Why?
 23. What relationships would you create?
 24. What cardinality would you expect?
 25. Why is a Star Schema useful?
-

PART F — Calculate

Tech With Shivansh

Build Projects • Get Hired



- 26. Which metrics should be Measures?
- 27. Which calculations might be Calculated Columns?
- 28. Where would DAX be useful?
- 29. How would you calculate Profit Margin?
- 30. How would you calculate YoY Growth?
- 31. How would you calculate YTD?
- 32. How would you handle filter context?

PART G — Analyze

- 33. How would you identify the best region?
- 34. How would you identify the worst region?
- 35. How would you investigate declining sales?
- 36. How would you identify profitable products?
- 37. How would you analyze customer performance?

PART H — Visualize

- 38. Which KPIs belong at the top?
- 39. Which chart would you use for trends?
- 40. Which chart would you use for regional comparison?
- 41. How would you avoid dashboard clutter?
- 42. How would you make the dashboard interactive?

PART I — Automate

- 43. Which repetitive tasks could VBA automate?
- 44. Would you use VBA for data cleaning if Power Query already handles it well?
- 45. How would you decide whether automation is worth building?

PART J — Business Storytelling

The interviewer finally asks:

"Your dashboard shows Sales ↑ 18%, but Profit ↓ 4%. What do you tell the CEO?"

- 46. What would you investigate?
- 47. What evidence would you need?
- 48. What could be the root causes?
- 49. How would you communicate the finding?
- 50. What action would you recommend?



THE FINAL STANDARD

If someone can solve **all 50 questions properly**, they shouldn't merely be able to say:



<https://www.linkedin.com/in/the-venom/>

Shivansh

"I know Excel."

They should be able to demonstrate:

DATA

Understand → Validate → Clean

TRANSFORM

Power Query → Repeatable ETL

MODEL

Fact → Dimension → Relationships → Star Schema

CALCULATE

Excel Formulas → DAX → KPIs

ANALYZE

Pivot Tables → Segmentation → Trends → Root Cause

VISUALIZE

Business Charts → Dashboard → Storytelling

AUTOMATE

Macros → VBA → Repetitive Workflows

COMMUNICATE

Finding → Insight → Recommendation → Business Decision

The ultimate interview question is therefore not:

"Which Excel function do you know?"

It's:

"Here is a messy business problem. Show me how you would turn it into a reliable analysis and a decision."

If a candidate can do that using the 1,000-row dataset—and explain **why** they selected each tool—they have moved well beyond basic Excel and into **job-ready Excel Data Analytics**.

